

Tutorial: exam-style practice

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Part A: 10 multiple-choice questions (1 mark each). Part B: 5 structured questions (27 marks). Same format as the final exam.

Answer Part A by circling one option. Answer Part B in full, showing your working. A periodic table and the answer key are at the back.

Part A · Multiple choice

A1 Topic: Solution SOL-LO1

Which pair of substances would form a solution?

- A. water and olive oil
- B. hexane and iodine
- C. water and hexane
- D. water and candle wax

A2 Topic: Solution SOL-LO1

A solution that contains less solute than its solubility limit at that temperature is _____.

- A. saturated
- B. supersaturated
- C. unsaturated
- D. immiscible

A3 Topic: Solution SOL-LO2

A 70% v/v alcohol hand rub contains how much ethanol in 250 mL?

- A. 70 mL
- B. 175 mL
- C. 25 mL
- D. 180 mL

A4 Topic: Solution SOL-LO2

What is the molarity of 4.00 g of NaOH (40.00 g/mol) in 250 mL of solution?

- A. 0.100 M
- B. 0.400 M
- C. 1.00 M
- D. 0.0160 M

A5 Topic: Solution SOL-LO2

Drinking water contains 0.20 mg of lead per litre. Its concentration is about _____.

- A. 0.20 ppm
- B. 0.20 ppb
- C. 200 ppm
- D. 20 ppm

A6 Topic: Solution SOL-LO3

50.0 mL of 0.80 M KCl is diluted to 200. mL. The new concentration is _____.

- A. 3.2 M
- B. 0.20 M
- C. 0.40 M
- D. 0.16 M

A7 Topic: Solution SOL-LO4

Which is a nonelectrolyte?

- A. NaCl
- B. KNO₃
- C. C₆H₁₂O₆
- D. MgCl₂

A8 Topic: Solution SOL-LO4

Which 0.1 M solution has the highest boiling point?

- A. glucose
- B. NaCl
- C. Al(NO₃)₃
- D. CaCl₂

A9 Topic: Solution SOL-LO5

The osmolarity of 0.15 M CaCl₂ is _____.

- A. 0.15 osmol
- B. 0.30 osmol
- C. 0.45 osmol
- D. 0.60 osmol

A10 Topic: Solution SOL-LO5

Red blood cells placed in a hypertonic solution will _____.

- A. swell and burst
- B. shrivel (crenation)
- C. stay the same
- D. dissolve

Part B • Structured questions

Show all working. The marks and Bloom's level are shown for each question.

Question B1 Topic: Solution SOL-L01 5 marks, C2

(a) Define the terms **solute**, **solvent** and **saturated solution**. (3 marks) (b) Iodine (I_2) is shaken with a mixture of water and hexane. Predict in which layer most of the iodine dissolves, and explain your answer. (2 marks)

Question B2 Topic: Solution SOL-L02 6 marks, C3

Normal saline is 0.9% m/v NaCl (Na = 22.99, Cl = 35.45). (a) Calculate the mass of NaCl in a 500 mL bag. (2 marks) (b) Calculate the molarity of normal saline. (2 marks) (c) A 1.5 L water sample contains 0.45 mg of lead. Calculate its concentration in ppm. (2 marks)

Question B3 Topic: Solution SOL-L02 5 marks, C3

An antacid contains calcium carbonate: $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$. Calculate the volume, in mL, of 0.250 M HCl that reacts with 1.00 g of $CaCO_3$ (Ca = 40.08, C = 12.01, O = 16.00).

Question B4 Topic: Solution SOL-L03 6 marks, C3

(a) 25.0 mL of 12.0 M HCl is diluted with water to 500. mL. Calculate the new concentration. (2 marks) (b) A pharmacist must prepare 200. mL of 0.50% m/v KCl from a 15% m/v stock solution. Calculate the volume of stock needed and describe how the solution is made. (3 marks) (c) Explain why the number of moles of solute does not change on dilution. (1 mark)

Question B5 Topic: Solution SOL-L05 5 marks, C3

Blood plasma has an osmolarity of about 0.30 osmol. (a) Calculate the osmolarity of 0.15 M NaCl and of 0.10 M CaCl₂. (2 marks) (b) State whether these solutions are isotonic, hypotonic or hypertonic to red blood cells. (1 mark) (c) Predict and explain what happens to red blood cells placed in 0.050 M NaCl. (2 marks)

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Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01 2.1																	2 He 4.00 —
3 Li 6.94 1.0	4 Be 9.01 1.5											5 B 10.81 2.0	6 C 12.01 2.5	7 N 14.01 3.0	8 O 16.00 3.5	9 F 19.00 4.0	10 Ne 20.18 —
11 Na 22.99 0.9	12 Mg 24.31 1.2											13 Al 26.98 1.5	14 Si 28.09 1.8	15 P 30.97 2.1	16 S 32.07 2.5	17 Cl 35.45 3.0	18 Ar 39.95 —
19 K 39.10 0.8	20 Ca 40.08 1.0	21 Sc 44.96 1.3	22 Ti 47.87 1.5	23 V 50.94 1.6	24 Cr 52.00 1.6	25 Mn 54.94 1.5	26 Fe 55.84 1.8	27 Co 58.93 1.9	28 Ni 58.69 1.9	29 Cu 63.55 1.9	30 Zn 65.38 1.6	31 Ga 69.72 1.6	32 Ge 72.63 1.8	33 As 74.92 2.0	34 Se 78.97 2.4	35 Br 79.90 2.8	36 Kr 83.80 —
37 Rb 85.47 0.8	38 Sr 87.62 1.0	39 Y 88.91 1.2	40 Zr 91.22 1.4	41 Nb 92.91 1.6	42 Mo 95.95 1.8	43 Tc [98] 1.9	44 Ru 101.07 2.2	45 Rh 102.91 2.2	46 Pd 106.42 2.2	47 Ag 107.87 1.9	48 Cd 112.41 1.7	49 In 114.82 1.7	50 Sn 118.71 1.8	51 Sb 121.76 1.9	52 Te 127.60 2.1	53 I 126.90 2.5	54 Xe 131.29 —
55 Cs 132.91 0.7	56 Ba 137.33 0.9	57–71	72 Hf 178.49 1.3	73 Ta 180.95 1.5	74 W 183.84 1.7	75 Re 186.21 1.9	76 Os 190.23 2.2	77 Ir 192.22 2.2	78 Pt 195.08 2.2	79 Au 196.97 2.4	80 Hg 200.59 1.9	81 Tl 204.38 1.8	82 Pb 207.2 1.9	83 Bi 208.98 1.9	84 Po [209] 2.0	85 At [210] 2.2	86 Rn [222] —
87 Fr [223] 0.7	88 Ra [226] 0.9	89–103	104 Rf [267] —	105 Db [268] —	106 Sg [269] —	107 Bh [270] —	108 Hs [277] —	109 Mt [278] —	110 Ds [281] —	111 Rg [282] —	112 Cn [285] —	113 Nh [286] —	114 Fl [289] —	115 Mc [290] —	116 Lv [293] —	117 Ts [294] —	118 Og [294] —
Lanthanides		57 La 138.91 —	58 Ce 140.12 —	59 Pr 140.91 —	60 Nd 144.24 —	61 Pm [145] —	62 Sm 150.36 —	63 Eu 151.96 —	64 Gd 157.25 —	65 Tb 158.93 —	66 Dy 162.50 —	67 Ho 164.93 —	68 Er 167.26 —	69 Tm 168.93 —	70 Yb 173.05 —	71 Lu 174.97 —	
Actinides		89 Ac [227] 1.1	90 Th 232.04 1.3	91 Pa 231.04 1.4	92 U 238.03 1.4	93 Np [237] —	94 Pu [244] —	95 Am [243] —	96 Cm [247] —	97 Bk [247] —	98 Cf [251] —	99 Es [252] —	100 Fm [257] —	101 Md [258] —	102 No [259] —	103 Lr [262] —	

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

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Part A: 10 multiple-choice questions (1 mark each). Part B: 5 structured questions (27 marks). Same format as the final exam.

Answer key

Part A · Multiple choice

A1 Topic: Solution SOL-LO1

Which pair of substances would form a solution?

- A. water and olive oil
- B. hexane and iodine**
- C. water and hexane
- D. water and candle wax

Both hexane and I_2 are nonpolar: like dissolves like.

A2 Topic: Solution SOL-LO1

A solution that contains less solute than its solubility limit at that temperature is _____.

- A. saturated
- B. supersaturated
- C. unsaturated**
- D. immiscible

Below the limit = unsaturated.

A3 Topic: Solution SOL-LO2

A 70% v/v alcohol hand rub contains how much ethanol in 250 mL?

- A. 70 mL
- B. 175 mL**
- C. 25 mL
- D. 180 mL

$$250 \text{ mL} \times (70 \div 100) = 175 \text{ mL.}$$

A4 Topic: Solution SOL-LO2

What is the molarity of 4.00 g of NaOH (40.00 g/mol) in 250 mL of solution?

- A. 0.100 M
- B. 0.400 M**
- C. 1.00 M
- D. 0.0160 M

$$0.100 \text{ mol} \div 0.250 \text{ L} = 0.400 \text{ M.}$$

A5 Topic: Solution SOL-LO2

Drinking water contains 0.20 mg of lead per litre. Its concentration is about _____.

- A. 0.20 ppm**
- B. 0.20 ppb
- C. 200 ppm
- D. 20 ppm

In water, $1 \text{ mg/L} \approx 1 \text{ ppm}$.

A6 Topic: Solution SOL-LO3

50.0 mL of 0.80 M KCl is diluted to 200. mL. The new concentration is _____.

- A. 3.2 M
- B. 0.20 M**
- C. 0.40 M
- D. 0.16 M

$$(0.80 \times 50.0) \div 200. = 0.20 \text{ M.}$$

A7 Topic: Solution SOL-LO4

Which is a nonelectrolyte?

- A. NaCl
- B. KNO₃
- C. C₆H₁₂O₆**
- D. MgCl₂

Glucose dissolves as neutral molecules.

A8 Topic: Solution SOL-LO4

Which 0.1 M solution has the highest boiling point?

- A. glucose
- B. NaCl
- C. Al(NO₃)₃**
- D. CaCl₂

Al(NO₃)₃ gives 4 particles per formula unit: the most particles.

A9 Topic: Solution SOL-LO5

The osmolarity of 0.15 M CaCl₂ is _____.

- A. 0.15 osmol
- B. 0.30 osmol
- C. 0.45 osmol**
- D. 0.60 osmol

$0.15 \times 3 = 0.45$ osmol.

A10 Topic: Solution SOL-LO5

Red blood cells placed in a hypertonic solution will _____.

- A. swell and burst
- B. shrivel (crenation)**
- C. stay the same
- D. dissolve

Water leaves the cells towards the higher osmolarity outside.

Part B · Structured questions

Question B1

Topic: Solution

SOL-L01

5 marks, C2

(a) Define the terms **solute**, **solvent** and **saturated solution**. (3 marks) (b) Iodine (I_2) is shaken with a mixture of water and hexane. Predict in which layer most of the iodine dissolves, and explain your answer. (2 marks)

(a) **Solute**: the minor component that is dissolved. **Solvent**: the major component that does the dissolving (usually the same phase as the solution). **Saturated solution**: a solution that holds the maximum amount of solute that can dissolve at that temperature. (b) In the **hexane** layer. I_2 and hexane are both nonpolar, with similar intermolecular forces (like dissolves like); water is polar.

Step	Marking criteria	Marks
1	Solute defined	1
2	Solvent defined	1
3	Saturated solution defined	1
4	Hexane layer	1
5	Both nonpolar: like dissolves like	1
	Total	5

Question B2

Topic: Solution

SOL-L02

6 marks, C3

Normal saline is 0.9% m/v NaCl (Na = 22.99, Cl = 35.45). (a) Calculate the mass of NaCl in a 500 mL bag. (2 marks) (b) Calculate the molarity of normal saline. (2 marks) (c) A 1.5 L water sample contains 0.45 mg of lead. Calculate its concentration in ppm. (2 marks)

(a) $500 \text{ mL} \times (0.9 \text{ g} \div 100 \text{ mL}) = \mathbf{4.5 \text{ g NaCl}}$. (b) $0.9\% = 9.0 \text{ g/L}$; $9.0 \text{ g} \div 58.44 \text{ g/mol} = 0.154 \text{ mol}$, so $\mathbf{0.154 \text{ M}}$. (c) $0.45 \text{ mg} \div 1.5 \text{ L} = 0.30 \text{ mg/L} = \mathbf{0.30 \text{ ppm}}$.

Step	Marking criteria	Marks
1	Correct use of 0.9 g per 100 mL	1
2	4.5 g NaCl	1
3	Moles in 1 L: $9.0 \div 58.44$	1
4	0.154 M	1
5	$0.45 \text{ mg} \div 1.5 \text{ L} = 0.30 \text{ mg/L}$	1
6	0.30 ppm	1
	Total	6

Question B3 Topic: Solution SOL-L02 5 marks, C3

An antacid contains calcium carbonate: $\text{CaCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{CaCl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$. Calculate the volume, in mL, of 0.250 M HCl that reacts with 1.00 g of CaCO_3 (Ca = 40.08, C = 12.01, O = 16.00).

$M(\text{CaCO}_3) = 40.08 + 12.01 + 3(16.00) = 100.09 \text{ g/mol}$. Moles = $1.00 \div 100.09 = 9.99 \times 10^{-3} \text{ mol}$. Moles HCl = $2 \times 9.99 \times 10^{-3} = 0.0200 \text{ mol}$. Volume = $0.0200 \div 0.250 = 0.0799 \text{ L} = \mathbf{79.9 \text{ mL}}$.

Step	Marking criteria	Marks
1	Molar mass 100.09 g/mol	1
2	$9.99 \times 10^{-3} \text{ mol CaCO}_3$	1
3	$\times 2$ mole ratio: 0.0200 mol HCl	1
4	0.0799 L	1
5	79.9 mL (accept 80 mL)	1
	Total	5

Question B4 Topic: Solution SOL-L03 6 marks, C3

(a) 25.0 mL of 12.0 M HCl is diluted with water to 500. mL. Calculate the new concentration. (2 marks) (b) A pharmacist must prepare 200. mL of 0.50% m/v KCl from a 15% m/v stock solution. Calculate the volume of stock needed and describe how the solution is made. (3 marks) (c) Explain why the number of moles of solute does not change on dilution. (1 mark)

(a) $M_2 = (12.0 \text{ M} \times 25.0 \text{ mL}) \div 500. \text{ mL} = \mathbf{0.600 \text{ M}}$. (b) $(15\%)V_1 = (0.50\%)(200. \text{ mL})$, $V_1 = \mathbf{6.67 \text{ mL}}$ of stock; measure it into a 200 mL volumetric flask and **make up to 200 mL** with water (about 193 mL of water). (c) Only solvent is added; no solute is added or removed, so the moles of solute stay the same.

Step	Marking criteria	Marks
1	Correct $M_1V_1 = M_2V_2$ set-up	1
2	0.600 M	1
3	Correct set-up for KCl	1
4	6.67 mL	1
5	Made up to 200 mL with water	1
6	Only solvent is added	1
	Total	6

Question B5 Topic: Solution SOL-L05 5 marks, C3

Blood plasma has an osmolarity of about 0.30 osmol. (a) Calculate the osmolarity of 0.15 M NaCl and of 0.10 M CaCl₂. (2 marks) (b) State whether these solutions are isotonic, hypotonic or hypertonic to red blood cells. (1 mark) (c) Predict and explain what happens to red blood cells placed in 0.050 M NaCl. (2 marks)

(a) NaCl: $0.15 \times 2 = \mathbf{0.30 \text{ osmol}}$. CaCl₂: $0.10 \times 3 = \mathbf{0.30 \text{ osmol}}$. (b) Both are **isotonic**. (c) $0.050 \times 2 = 0.10 \text{ osmol}$, lower than inside the cells: **hypotonic**. Water moves into the cells by osmosis; they swell and burst (**haemolysis**).

Step	Marking criteria	Marks
1	0.30 osmol (NaCl)	1
2	0.30 osmol (CaCl ₂)	1
3	Both isotonic	1
4	0.10 osmol: hypotonic	1
5	Water enters, cells swell and burst (haemolysis)	1
	Total	5

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